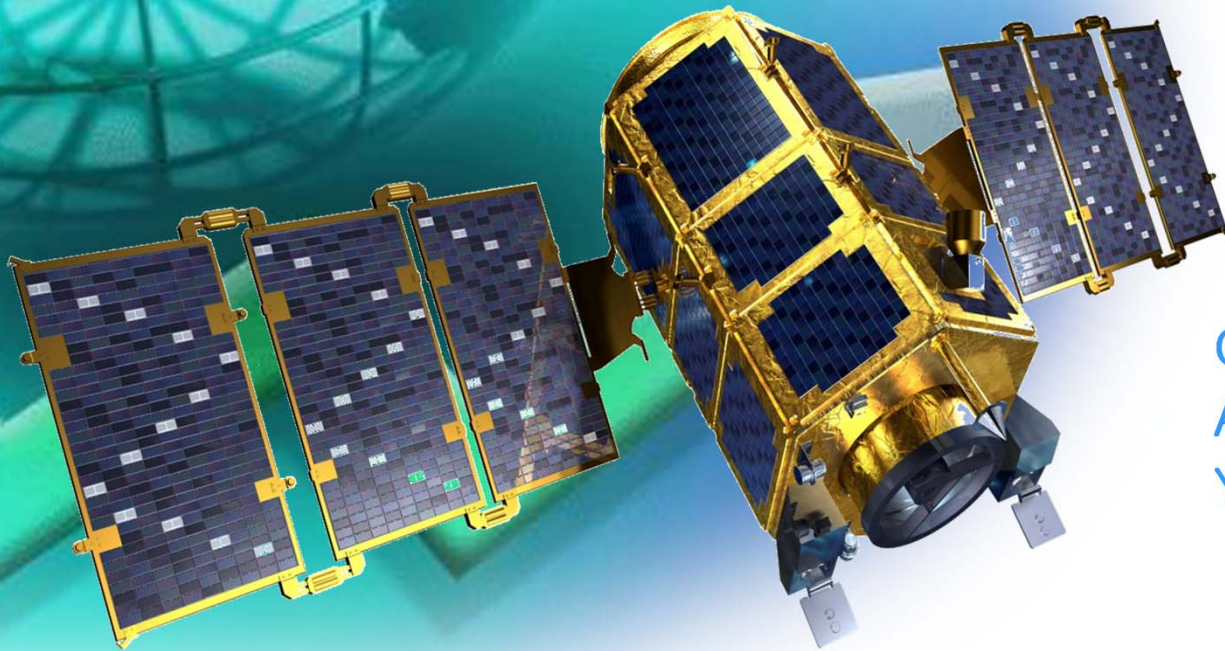
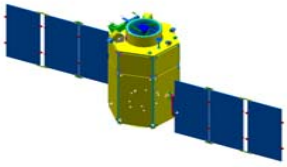


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Spacecraft Systems



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Yonsei University



Chapter 10

SPACECRAFT DESIGN & SIZING

10.1 Requirements, Constraints, and the Design Process

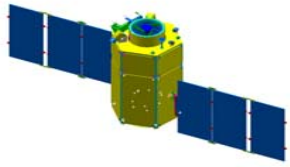
10.2 Spacecraft Configuration

10.3 Design Budgets

10.4 Designing the Spacecraft Bus

10.5 Integrating the Spacecraft Design

10.6 Examples

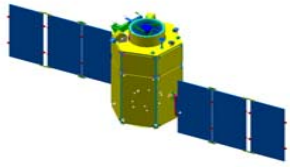


10.3 Design Budgets

◆ Overview of S/C Design and Sizing (Table 10–1)

TABLE 10-1. Overview of Spacecraft Design and Sizing. The process is highly iterative, normally requiring several cycles through the table even for preliminary designs.

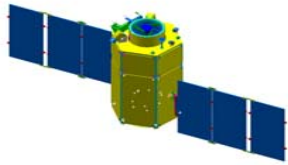
Step	References
1. Prepare list of design requirements and constraints	Sec. 10.1
2. Select preliminary spacecraft design approach and overall configuration based on the above list	Sec. 10.2
3. Establish budgets for spacecraft propellant, power, and weight	Sec. 10.3
4. Develop preliminary subsystem designs	Sec. 10.4
5. Develop baseline spacecraft configuration	Secs. 10.4,10.5
6. Iterate, negotiate, and update requirements, constraints, design budgets	Steps 1 to 5



10.3 Design Budgets

- ◆ Allocating Performance begins by establishing budgets (allocation) for:
 - Propellant
 - Power
 - Weight
 - Reliability

- ◆ Budget
 - Numerical list of the components of any overall system parameter
 - Page 4 of Chapter 1

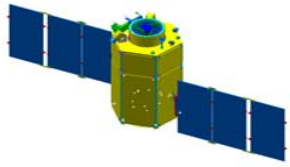


10.3 Design Budgets

- ◆ Propellant Budget can be derived by estimating the propellant requirements for:
 - Velocity (or Orbit) Correction and Control (ΔV)
 - Attitude Control

TABLE 10-7. Propellant Budget.

Elements	Reference
<i>Velocity Correction and Control</i>	Eqs. (17-9), (7-14), Table 7-3
<i>Attitude Control</i>	Secs. 10.4.2, 11.1
Spinup and despin	Eq. (10-9)
Maneuvering while spinning	Eq. (10-11)
Cancelling disturbance torque	Table 10-18
Control during ΔV thrusting	Table 10-18 and Eqs. (10-8a,b)
Attitude maneuvering	Eq. (10-3)
Limit cycling	Eqs. (10-5) through (10-7)
<i>Nominal Propellant</i>	Sum of above
<i>Margin</i>	10%–25% of nominal
<i>Residual</i>	1%–2% of total
<i>Total Propellant</i>	Sum of above



10.3 Design Budgets

◆ Velocity Correction & Control (ΔV)

- 1st/2nd Burn (for Coplanar Circular Hohmann Transfer)

$$\begin{aligned}\Delta V_1 &= \sqrt{2\mu / r_1 - \mu / a_{trans}} - \sqrt{\mu / r_1} \\ \Delta V_2 &= \sqrt{2\mu / r_2 - \mu / a_{trans}} - \sqrt{\mu / r_2} \\ \Delta V &= \Delta V_1 + \Delta V_2\end{aligned}\quad (6-32)$$

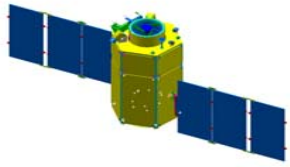
- Orbit Plane Change

- Simple Circular Plane Change (Plane Only!)

$$\Delta V = 2v_i \sin(\Delta i / 2), \quad v_i = v_f$$

- Combined Plane Change (Plane & Velocity)

$$\Delta V = \sqrt{v_i^2 + v_f^2 - 2v_i v_f \cos(\Delta i)}, \quad v_i \neq v_f$$



10.3 Design Budgets

◆ Velocity Correction & Control (ΔV) cont'd.....

■ Altitude Maintenance (LEO)

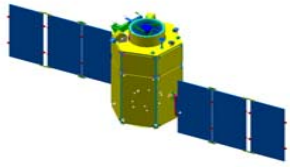
$$\Delta V_A = \pi(C_b A / m) \rho a V \quad (\text{per revolution}) \quad (6-26)$$

C_b : drag coefficient, A : S/C crosssectional area, m : S / C mass

ρ : atmospheric density, a : semimajor axis, V : S/C velocity

$m / C_b A$: ballistic coefficient (constant for most S/C)

- due to Atmospheric Drag
- valid for near circular orbit



10.3 Design Budgets

◆ Velocity Correction & Control (ΔV) cont'd.....

■ North/South Station Keeping (GEO)

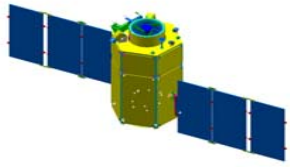
$$\Delta V_{MOON} = 102.67 \cos \alpha \sin \alpha \text{ (m / s / year)} \quad (6-51)$$

α : angle between orbital plane and Moon's orbit

$$\Delta V_{SUN} = 40.17 \cos \gamma \sin \gamma \text{ (m / s / year)} \quad (6-51)$$

γ : angle between orbital plane and ecliptic plane

- Necessary when mission requirements limit inclination drift due to Earth's non-spherical geopotential & 3rd body.



10.3 Design Budgets

◆ Velocity Correction & Control (ΔV) cont'd.....

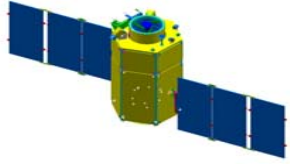
■ East/West Station Keeping (GEO)

$$\Delta V_{EW} = 1.715 \sin(2 |l_D - l_S|) \quad (m/s/year) \quad (6-53)$$

l_D : desired longitude

l_S : closest stable longitude ($E75^\circ, 255^\circ$)

- Change in velocity required to compensate for the drift (east–west) and to maintain the satellite in the vicinity of the specified longitude
- S/C tends to drift the closest stable longitude due to the Earth's oblateness



10.3 Design Budgets

◆ Velocity Correction & Control (ΔV) cont'd.....

■ Rephasing/Rendezvous

$$\Delta V_R = \frac{1}{1080} v \cdot (\text{drift rate}) \quad (6-41)$$

v : orbital velocity, drift rate: deg/orbit for relative phasing

■ S/C disposal (maneuver from r to h in altitude, $r > h$)

$$\Delta V_{ORBIT} \approx v \left[1 - \sqrt{2R_E / (R_E + h_{DEORBIT} + r)} \right] \quad (6-54)$$

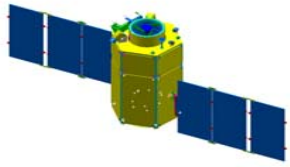
R_E : radius of Earth, $h_{DEORBIT}$: target altitude

■ Total ΔV is equal to the sum of the above

■ Propellant Mass

$$m_f = m_0 e^{-(\Delta V/v_0)}$$

$$m_p = m_0 - m_f$$



10.3 Design Budgets

◆ Attitude Control

- Spinup & Despin (in case of Spin-Stabilized S/C)
 - Angular Impulse required for Spinup in $N \cdot m \cdot s$

$$L_{SPINUP} = I_{SS} \Omega_S \quad (10-9)$$

I_{SS} : S/C moment of Inertia, Ω_S : spin speed in rad / sec

- Associated Propellant Mass

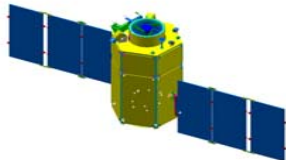
$$m_p = L / s I_{SP} g, \quad (10-8b)$$

s : thruster lever arm, I_{SP} : specific impulse, g : gravity acceleration

- Maneuvering while Spinning
 - Spin-axis re-orientation by α radians requires us to precess the angular momentum vector.

$$L_{SPINUP} = I_{SS} \Omega_S \alpha \quad (10-11)$$

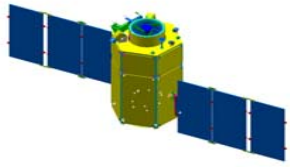
- Cancelling Disturbance Torques



10.3 Design Budgets

TABLE 10-18. Disturbance Torques. These are vector equations where “ $\times$ ” denotes vector cross product and “ $\cdot$ ” denotes vector dot product. See Sec. 11.1 for simplified equations.

Disturbance	Equation	Definition of Terms
ΔV Thruster Misalignment	$\mathbf{s} \times \mathbf{T}$	$\mathbf{s}$ vector distance from center of mass to thrust application point $\mathbf{T}$ vector thrust
Aerodynamic Torque	$\frac{1}{2} \rho V^2 C_d A (\mathbf{u}_v \times \mathbf{s}_{cp})$	ρ atmospheric density C_d drag coefficient (typically 2.25) A area perpendicular to $\mathbf{u}_v$ V velocity $\mathbf{u}_v$ unit vector in velocity direction $\mathbf{s}_{cp}$ vector distance from center of mass to center of pressure
Gravity Gradient Torque	$\frac{3\mu}{R_0^3} \mathbf{u}_e \times (\mathbf{I} \cdot \mathbf{u}_e)$	μ Earth's gravitational coefficient $3.986 \times 10^{14} \text{ m}^3/\text{s}^2$ R_0 Distance to Earth's center (m) $\mathbf{I}$ Spacecraft inertia tensor $\mathbf{u}_e$ Unit vector toward nadir
Solar Radiation Torque	$K_s (\mathbf{u}_s \cdot \mathbf{u}_n) A \left[\mathbf{u}_s (\alpha + r_d) + \mathbf{u}_n \left\{ 2r_s + \frac{2}{3} r_d \right\} \right] \times \mathbf{s}_c$	K_s solar pressure constant $4.644 \times 10^{-6} \text{ N/m}^2$ $\mathbf{s}_c$ vector from spacecraft center of mass to area A $\mathbf{u}_n$ unit vector normal to A $\mathbf{u}_s$ unit vector toward the Sun α surface absorptivity coefficient r_s surface specular reflectance coefficient r_d surface diffuse reflectance coefficient (Note: $\alpha + r_s + r_d = 1$)



10.3 Design Budgets

◆ Attitude Control cont'd.....

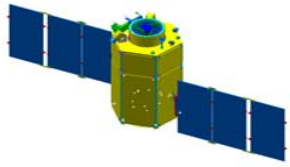
- Attitude Maneuvering
 - Angular Impulse required to start an attitude maneuver

$$L_{START} = I_{SS} \omega_{MAN} \quad (10-3)$$

I_{SS} : S/C moment of Inertia, ω_{MAN} : angular rate of the maneuver

- Limit Cycling
 - Thruster control systems operate by pulsing a thruster when the attitude error exceed a set value known as dead-zone limit
 - Total Angular Impulse during the Mission

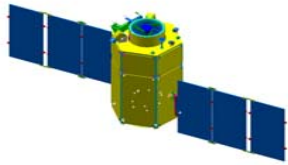
$$L_M = (\text{Average Impulse Rate}) \times (\text{Mission Duration}) \quad (10-7)$$



10.3 Design Budgets

◆ Propellant Budget = Nominal + Margin + Residual

- Nominal Propellant is the sum of propellants for:
 - Velocity Correction and Control (ΔV)
 - Attitude Control
- Margin: 10%~25% of Nominal
- Residual (Unavailable): 1%~2% of (Nominal + Margin)



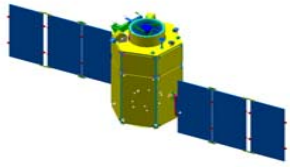
10.3 Design Budgets

◆ Power Budget

- Estimate power requirements for the payload and the S/C subsystems

TABLE 10-8. Steps in Preparing a Power Budget.

Step	What's Involved	Where Discussed
1. Prepare Operating Power Budget	Estimate power requirements for payload and each spacecraft bus subsystem	Chaps. 9, 11, 13
2. Size the Battery	Estimate power level that the battery must supply Compute discharge cycle duration, charge cycle duration, and number of charge-discharge cycles Select depth of discharge Select charge rate Compute battery recharge power	Generally equal to or less than the operating power level Determined by orbit selection and mission duration (Chap. 7) Sec.11.4 Sec.11.4 Sec.11.4
3. Estimate Power Degradation Over Mission Life	Compute degradation of power system from orbital environment	Secs. 8.1, 11.4

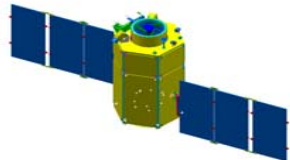


10.3 Design Budgets

- Step 1: Prepare Operating Power Budget
 - Budget separately for each operating mode
 - Pay particular attention to peak power needs for each sub system

- Step 2: Size the Battery

- Step 3: Est. Power Degradation over Mission Life
 - Example: 30% degradation for 10yrs GEO

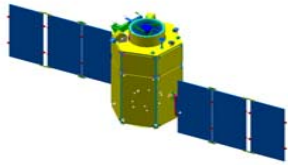


10.3 Design Budgets

TABLE 10-9. Typical Power Consumption by Module or Subsystem.

Spacecraft Size Spacecraft Power	Minimum (< 100 W total)	% of Operating Power		References
		Small (~ 200 W)	Medium to Large (> 500 W)	
<i>Subsystem*</i>				
Payload	20–50 W	40	40–80	Chaps. 9, 13
Propulsion	0	0	0–5	Sec. 10.4.1, Chap. 17
Attitude Control	0	15	5–10	Secs. 10.4.2, 11.1
Communications	15W	5	5–10	Secs. 10.4.3, 11.2
Command & Data Handling	5 W	5	5–10	Secs. 10.4.4, 11.3
Thermal	0	5	0–5	Secs. 10.4.5, 11.5
Power	10–30 W	30	5–25	Secs. 10.4.6, 11.4
Structure	0	0	0	Secs. 10.4.7, 11.6
<i>Average Power: Sum of above</i>				
<i>Margin: 5% to 25% of power based on design maturity</i>				
<i>Total Average Requirement for Operating Power: Total of above</i>				

* Includes conversion and line losses.

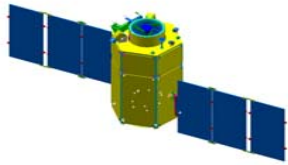


10.3 Design Budgets

◆ Weight Budget

TABLE 10-10. Weight Budget. The percentages shown in the right-hand column are the percent of spacecraft dry weight.

Element	Weight	Reference	Comments
<i>Payload</i>	$M_{P/L}$	Chaps. 9, 13	15% to 50% of M_{dry}
<i>Spacecraft Subsystems</i>	M_{SS}		Sum of subsystem wts
Propulsion	$M_{propulsion}$	Chap. 17	
Attitude Control	M_{gnc}	Sec. 11.1	
Communications	M_{com}	Sec. 11.2	
Command & Data Handling	M_{cadh}	Sec. 11.3	
Thermal	M_{th}	Sec. 11.5	2% to 5% of M_{dry}
Power	M_{ep}	Sec. 11.4	
Structure & Mechanisms	M_{sam}	Sec. 11.6	8% to 12% of M_{inj} or 15% to 25% of M_{dry}
<i>Margin</i>	M_{mar}		5% to 25% of wt based on design maturity
<i>Spacecraft Dry Weight</i>	$M_{dry} = M_{P/L} + M_{SS} + M_{mar}$		
<i>Propellant</i>	M_{prop}	Table 10-7	
<i>Loaded Weight</i>	$M_{loaded} = M_{dry} + M_{prop}$	Chap. 17	
<i>Kick Stage</i>	M_{kick}	Sec. 17.3	
<i>Injected Weight</i>	$M_{inj} = M_{loaded} + M_{kick}$		
<i>Adapter</i>	$M_{adapter}$	Sec. 11.6	
<i>Boosted Weight</i>	$M_{boosted} = M_{inj} + M_{adapter}$	Chap. 18	



10.3 Design Budgets

◆ Reliability Budget

TABLE 10-11. Preparing a Reliability Budget.

Step	Comments	Reference																
1. Establish mission success criteria	The criteria should be numerical and may have multiple elements. For instance, a communication spacecraft having multiple channels for several types of service might define success as one channel of each type of service or as a total number of channels and total radiated power.	Chaps. 1, 2, 3																
2. Assign numerical success probability to each element of the mission success criteria and define the method for computing success probability	This might be stated as a probability of 0.5 of operating service "A" for 5 years and a probability of 0.7 of operating service "B" for 2 years and 0.4 of operating service "B" for 7 years. For each element of the success criteria, numerical values and associated lifetimes are assigned. Several methods of evaluating success probability are available—see Chap. 19.	Sec. 19.2																
3. Create the reliability budget by allocating the success probability (reliability) to each item of hardware & software	If, for instance, a system reliability of 0.6 is required, it might be allocated as: <table><tr><td>Propulsion</td><td>0.95</td><td>Comm</td><td>0.93</td></tr><tr><td>Structure</td><td>0.99</td><td>C&DH</td><td>0.93</td></tr><tr><td>Thermal</td><td>0.99</td><td>Power</td><td>0.93</td></tr><tr><td>ADCS</td><td>0.9</td><td>Payload</td><td>0.89</td></tr></table>	Propulsion	0.95	Comm	0.93	Structure	0.99	C&DH	0.93	Thermal	0.99	Power	0.93	ADCS	0.9	Payload	0.89	Sec. 19.2
Propulsion	0.95	Comm	0.93															
Structure	0.99	C&DH	0.93															
Thermal	0.99	Power	0.93															
ADCS	0.9	Payload	0.89															
4. Evaluate the system reliability and iterate the design to maximize reliability and identify and eliminate failure modes	Assuming independent, serial operation, hardware failure rate is generally evaluated by summing piece part failure rates. (See Chap. 19 and MIL-HDBK-17 [1991].) Failure mode analysis and elimination are discussed in Sec. 10.4. Effect of failures can be reduced and reliability raised by changing the design, selecting more reliable hardware, or adding redundant hardware and software.	Sec. 19.2																